

GC9A01 display is mounted on top of the board for clear visibility.

The soldering process begins with preparing the IndusBoard Coin V2. The required pins are identified using the labelled diagram. These include pins such as 3V, GND, TX, RX, and digital pins 1 through 44. The board should be cleaned to avoid soldering issues.

The RDA5807M FM module and CST816S touch controller are soldered onto the back of the board to save space.

For the RDA5807M FM module, pin 5 (SCLK) is connected to IndusBoard pin 5 (SCL), while pin 4 (SDIO) connects to pin 6 (SDA). Pin 2 (GND) is linked to any available ground pin, such as pin 1, and pin 7 (VDD) is connected to a 3V pin, for instance, pin 2. Pins 9 (LOUT) and 10 (ROUT) are soldered directly to the speakers (SP1 and SP2) to enable stereo audio output. A small antenna wire is attached to pin 3 (FMIN) and secured in place to ensure effective signal reception.

For the CST816S touch controller, pin 6 (SCL) is soldered to IndusBoard pin 5, and pin 5 (SDA) is connected to pin 6, sharing the I²C bus with the FM module. Pin 4 (RST) is linked to pin 4, and pin 3 (INT) is connected to pin 3 on the IndusBoard. Ground and power are supplied by connecting pin 2 (GND) to a ground pin and pin 1 (VDD) to a 3V pin. Fig. 6 shows the components soldered with IndusBoard.

FPC integration

The IndusBoard Coin also supports and hosts its own FPC ecosystem. Custom modules can be designed to integrate seamlessly using the FPC bus. For example, a custom FM radio module can be developed to connect to the IndusBoard Coin via the FPC connector. Fig. 7 shows the double-sided PCB of the project, and Fig. 8 shows the IndusBoard with an FPC connector.

Testing

To test the device, power it using a 3.3V battery and perform the following checks:

Change the FM station using the buttons. The interface displays three touch-sensitive buttons:

- Previous button (left, at coordinates prevButtonX = 60, prevButtonY = 150): A grey circle with a left triangle and a thin rectangle
- Next button (right, at coordinates nextButtonX = 180, nextButtonY = 150): A grey circle with a right triangle and a thin rectangle
- Test the previous button:
 - Tapping this button decreases the frequency by 0.1MHz (for example, from 87.5MHz to 87.4MHz)
 - If the frequency falls below 87.5MHz, it wraps



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